

**(a)** Complete Table 6.1 to show the correct order of stages in the mitotic cell cycle.

### Table 6.1

| stage of cell cycle                                                                |                      |           |
|------------------------------------------------------------------------------------|----------------------|-----------|
|  | G <sub>1</sub> phase | }         |
|                                                                                    | .....                |           |
|                                                                                    | .....                |           |
|                                                                                    | .....                | } mitosis |
|                                                                                    | .....                |           |
|                                                                                    | .....                |           |
|                                                                                    | telophase            |           |
|                                                                                    | cytokinesis          |           |

[3]

At various points during the mitotic cell cycle, checks are made. A cell goes through cell death (apoptosis) if errors occur that cannot be repaired. This makes sure that the daughter cells produced are genetically identical to each other and to the original cell.

Drugs have been developed that can inhibit the mitotic cell cycle and cause the cell to carry out apoptosis. These drugs are used in the treatment of cancer.

**(b)** Vincristine and 5-fluorouracil are chemical compounds that act as cell cycle inhibitors and can lead to apoptosis.

- Vincristine binds to spindle microtubules and prevents the spindle from carrying out its function.
- 5-fluorouracil prevents the synthesis of thymine nucleotides.

Complete Table 6.2 to show which event in the cell cycle will occur when Vincristine or 5-fluorouracil are added to healthy dividing cells at the start of the interphase stage of the cell cycle.

Place a tick (✓) if the event will occur or a cross (✗) if the event will **not** occur.

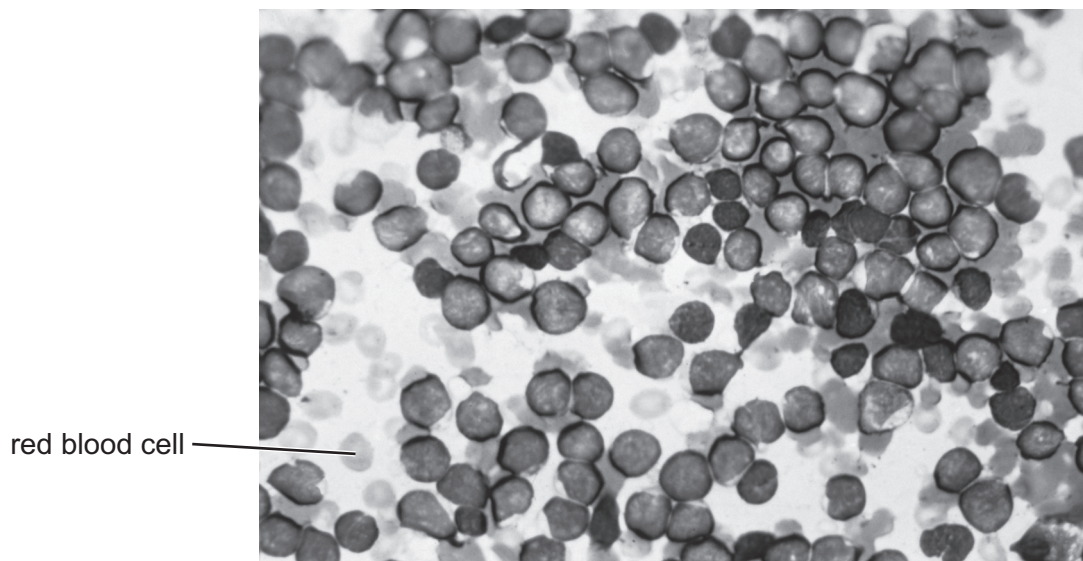
All boxes in the table should be completed.

**Table 6.2**

| <div>event in cell cycle</div> <div>compound</div> | S-phase completes | cell enters prophase of mitosis | chromosomes line up at spindle equator | sister chromatids move towards opposite poles |
|----------------------------------------------------|-------------------|---------------------------------|----------------------------------------|-----------------------------------------------|
| Vincristine                                        |                   |                                 |                                        |                                               |
| 5-fluorouracil                                     |                   |                                 |                                        |                                               |

- (c) Vincristine has been used in the treatment of certain types of leukaemia.

Fig. 6.1 is a photomicrograph of a blood smear of a person with one form of leukaemia, which affects lymphocytes.



**Fig. 6.1**

Describe how:

- the blood smear shown in Fig. 6.1 differs from a blood smear of a healthy person
- the lymphocytes in a person with leukaemia, such as those shown in Fig. 6.1, differ from those of a healthy person.

blood smear differences .....

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lymphocyte differences .....

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..... [3]

[Total: 8]